```
--- Part 1: Building a Simple Feedforward Neural Network ---

Original FNN training data shape: (60000, 28, 28)
Original FNN test data shape: (10000, 28, 28)
Flattened & Normalized FNN training data shape: (60000, 784)
Flattened & Normalized FNN test data shape: (10000, 784)

--- FNN Model Summary ---
/usr/local/lib/python3.13/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)

--- Training FNN Model ---

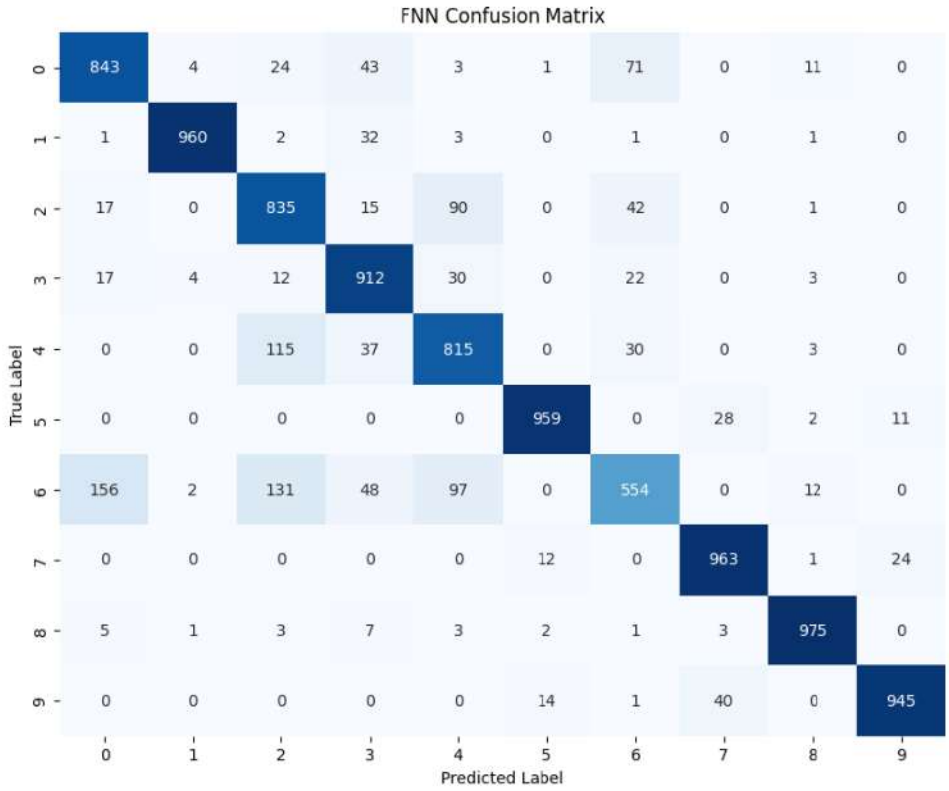
--- Evaluating FNN Model ---
FNN Test Loss: 0.3508
FNN Test Accuracy: 0.8761

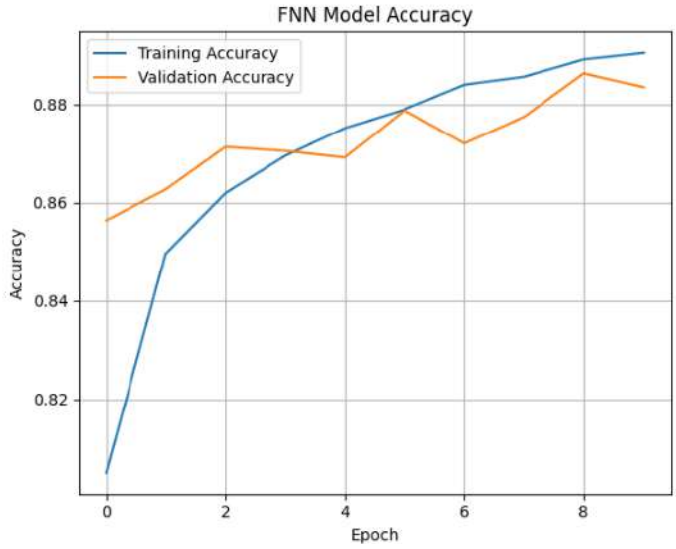
--- FNN Classification Report ---
              precision    recall  f1-score   support

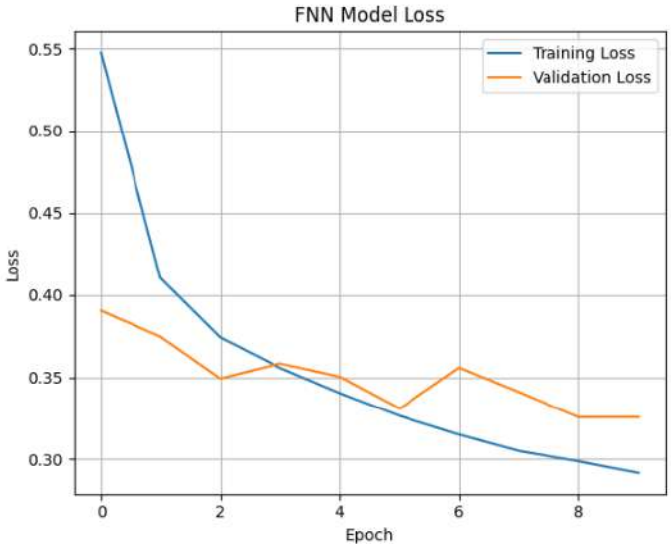
           0       0.81      0.84      0.83      1000
           1       0.99      0.96      0.97      1000
           2       0.74      0.83      0.79      1000
           3       0.83      0.91      0.87      1000
           4       0.78      0.81      0.80      1000
           5       0.97      0.96      0.96      1000
           6       0.77      0.55      0.64      1000
           7       0.93      0.96      0.95      1000
           8       0.97      0.97      0.97      1000
           9       0.96      0.94      0.95      1000

    accuracy                           0.88     10000
   macro avg       0.88      0.88      0.87     10000
weighted avg       0.88      0.88      0.87     10000
```

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--- FNN Confusion Matrix ---
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--- Part 2: Implementing a CNN ---

Original CNN training data shape: (60000, 28, 28)
Original CNN test data shape: (10000, 28, 28)
Reshaped & Normalized CNN training data shape: (60000, 28, 28, 1)
Reshaped & Normalized CNN test data shape: (10000, 28, 28, 1)

--- CNN Model Summary ---
/usr/local/lib/python3.13/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)

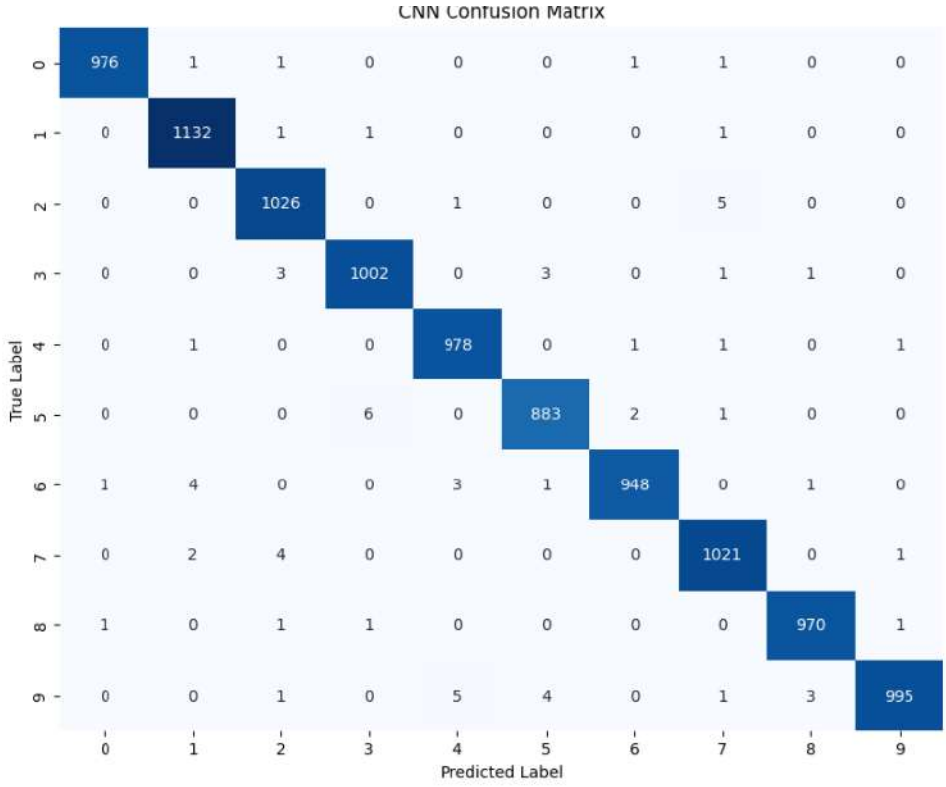
--- Training CNN Model ---

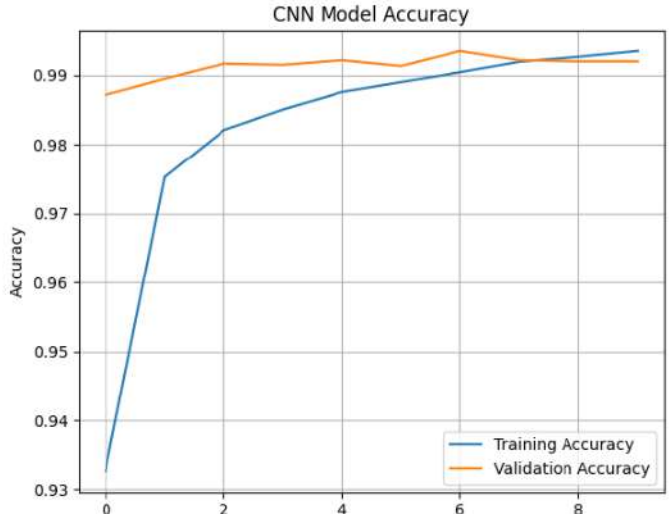
--- Evaluating CNN Model ---
CNN Test Loss: 0.0238
CNN Test Accuracy: 0.9951

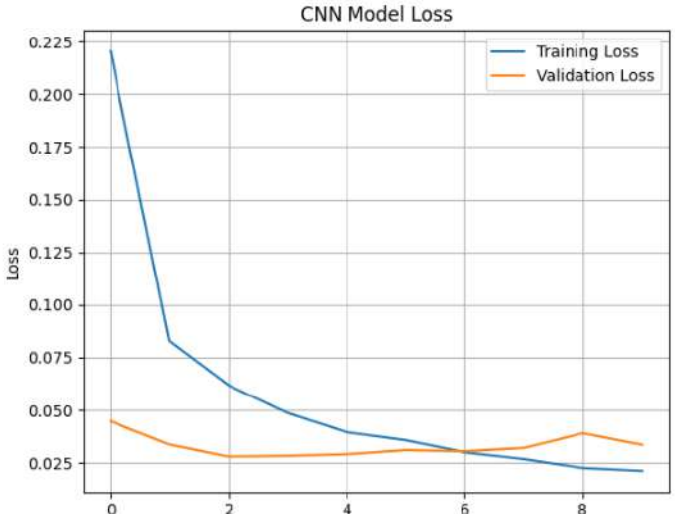
--- CNN Classification Report ---
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       980
           1       0.99      1.00      1.00      1135
           2       0.99      0.99      0.99      1032
           3       0.99      0.99      0.99      1010
           4       0.99      1.00      0.99       982
           5       0.99      0.99      0.99       892
           6       1.00      0.99      0.99       958
           7       0.99      0.99      0.99      1028
           8       0.99      1.00      1.00       974
           9       1.00      0.98      0.99      1009

    accuracy                           0.99     10000
   macro avg       0.99      0.99      0.99     10000
weighted avg       0.99      0.99      0.99     10000
```


CNN Confusion Matrix
True Label
Predicted Label
0 1 2 3 4 5 6 7 8 9
976 1 1 0 0 0 1 1 0 0
0 1132 1 1 0 0 0 1 0 0
0 0 1026 0 1 0 0 5 0 0
0 0 3 1002 0 3 0 1 1 0
0 1 0 0 978 0 1 1 0 1
0 0 0 6 0 883 2 1 0 0
1 4 0 0 3 1 948 0 1 0
0 2 4 0 0 0 0 1021 0 1
1 0 1 1 0 0 0 0 970 1
0 0 1 0 5 4 0 1 3 995


CNN Model Accuracy
Accuracy
Training Accuracy
Validation Accuracy


CNN Model Loss
Loss
Training Loss
Validation Loss

```
--- Sample CNN Predictions ---
```

